

VAMSI KRISHNA KOLLIPARA

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SUMMARY

Full-stack developer who has shipped healthcare web portals, RAG pipelines, and data automation systems using React, Next.js, Node.js, and PostgreSQL. MS in Computer Science (Indiana State University, GPA 3.66) with enterprise experience at Cognizant and two production RAG systems built with LangChain, OpenAI API, and Pinecone.

SKILLS

Languages: JavaScript, TypeScript, Python, SQL, HTML, CSS

Frontend: React, Next.js, Tailwind CSS

Backend: Node.js, Express.js, GraphQL, REST APIs

Databases: PostgreSQL, Pinecone

Cloud & DevOps: Azure, AWS, Docker, GitHub Actions, CI/CD, Vercel, Supabase

AI & LLM: LangChain, OpenAI API, Vector Databases, Retrieval-Augmented Generation

Tools: Git, Postman, Jest, React Testing Library, Pytest

RELEVANT EXPERIENCE

Full-Stack Developer, AIDM — Software Consultancy | Remote Jul 2025 – Present

- Built two healthcare web portals with Next.js (TypeScript), Node.js, GraphQL, and PostgreSQL, supporting multi-role access for clinical and administrative users; developed within a 6-person cross-functional team (3 developers, designer, project manager, IT specialist).
- Implemented security controls aligned with HIPAA requirements — AES-256 encryption for PHI at rest, TLS in transit, role-based access control (RBAC) with minimum-necessary access, and audit logging for sensitive user actions.
- Integrated Azure Active Directory with Microsoft Graph API for enterprise SSO, eliminating manual onboarding steps and automating email notification workflows across client organizations.
- Built a healthcare-domain RAG assistant (LangChain, OpenAI API, Pinecone) with PHI-aware document filtering, MMR-based retrieval, automated fallback to deterministic responses, and structured per-request logging for compliance auditing.
- Automated 5 internal approval workflows, reporting pipelines, and content publishing processes using n8n and Power Automate, eliminating ~10 hours/week of recurring manual effort.
- Set up PR-based code review workflows and GitHub Actions CI/CD pipelines with automated linting and type-checking gates; added structured JSON logging across AI pipeline stages for per-request tracing.

Graduate Research & Teaching Assistant, Indiana State University | Terre Haute, IN Aug 2023 – May 2025

- Developed and deployed a RAG chatbot on the university portal (LangChain, OpenAI API, Pinecone) to answer student queries on courses, deadlines, and academic policies; implemented configurable semantic retrieval with cosine similarity thresholds, chunk-overlap tuning, and vector indexing.
- Built and open-sourced WebWeaver, a Python crawling library, and used it to ingest 40,000+ university URLs into a curated knowledge base of ~8,000 documents powering the RAG chatbot's semantic search.

Full-Stack Developer, Cognizant Technology Solutions | Hyderabad, India Jul 2022 – Jul 2023

- Built and shipped React UI components and Express.js REST APIs for a client-facing logistics dashboard, delivering real-time shipment tracking and reporting features used by operations teams across multiple regions.
- Optimized SQL queries on multi-million-row databases — reduced average execution time by 30% through composite indexing, join restructuring, and eliminating N+1 query patterns across backend services.

Full-Stack Developer Intern, Cognizant Technology Solutions | Hyderabad, India Jan 2022 – May 2022

- Built frontend interfaces and integrated REST APIs for an internal reporting tool using JavaScript, HTML/CSS, and Node.js; developed SQL-driven dashboards delivering actionable insights to business stakeholders.

EDUCATION

Master of Science in Computer Science Aug 2023 – May 2025

Indiana State University, Terre Haute, IN | GPA: 3.66

Bachelor of Technology in Electronics & Communication Engineering Jun 2018 – May 2022

Vel Tech University, Chennai, India | GPA: 9.01/10

PROJECTS & OPEN-SOURCE

GitBar — macOS Menubar Git Dashboard (PyPI) Mar 2026

- Built and published an open-source macOS menubar app providing a unified Git dashboard aggregating PRs, issues, CI/CD status, and local repo health across GitHub, GitLab, and Bitbucket. Tech: Python, PyObjC, GitHub/GitLab REST APIs.

GadgetBox — Cross-Platform Developer Utilities (PyPI) Feb 2026

- Published a cross-platform system tray app bundling 12 developer utilities (JSON formatter, JWT decoder, UUID generator, Base64, hash, regex tester, etc.) with intelligent clipboard auto-detection. Tech: Python, pystray, tkinter.

WebWeaver — Web Scraping & Crawling Library (PyPI) Sep 2024

- Open-source Python library for configurable web crawling with URL validation, deduplication, robots.txt compliance, and recursive/breadth-first modes; powers the data ingestion layer behind the ISU RAG chatbot. Tech: Python, asyncio, aiohttp.